

Assignment-4
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- 1) ALU control unit takes Instruction [30,14-12] bit as input because this bit determines which ALU operation to do, like add, subtract, AND, OR Shift etc.
- 2) Zero pin indicates whether the result of ALU operation is zero or not. It is used for conditional branching like BEQ. It performs subtraction ($rs2 - rs1$) if the result is zero the zero pin is set to 1 and branching takes place, no separate compare instruction is needed.
- 3) For instruction,
Add x21, x22, x23
Single-cycle setup = $20 + 20 + 20 + 40 = 100s$
Pipelined setup = $5 \times 40 = 200s$
For this one instruction single-cycle setup will run faster
- 4) Ld x5, 10(x21)
 - i) ALU src : 1
 - ii) Reg Write : 1
 - iii) Mem Read : 1
 - iv) Mem Write : 0
 - v) Mem to Reg : 1